

Uncertainty-Aware Bayesian Mineral Prospectivity Mapping under Spatial Separation: Heterogeneous Predictor Relationships across the Central African Copperbelt

Abstract

Mineral prospectivity mapping integrates geological, geochemical, and geophysical evidence to identify locations whose mapped characteristics are associated with known deposits. We evaluate a partially pooled hierarchical Bayesian logistic model across four geographically contiguous held-out sectors of the Central African Copperbelt. The 1,872-cell modeling frame contains 138 documented occurrences and predictors for distance to faults, distance to lithological contacts, Bouguer gravity, and retained host-lithology classes. Intercepts and the four distance coefficients vary by Daly domain; gravity and retained host-rock effects are global. Posterior draws quantify parameter uncertainty, while each held-out cell receives the mean of its draw-wise conditional probabilities. The reported OOF artifacts do not include cell-level probability intervals or Bernoulli outcome draws. Predictor contributions to the decomposed OOF linear predictor vary across folds, and across four distance-representation specifications mean ROC-AUC ranges from 0.671 to 0.686 while mean PR-AUC ranges from 0.107 to 0.145. Distance associations recur across evaluated representations, whereas quadratic curvature is representation-dependent and population-level D^* is not identifiable. Bootstrap intervals quantify uncertainty in performance metrics, not individual cell probabilities. The results show geographic variation in fitted associations and held-out discrimination, but do not establish distinct causal geological processes or a universal distance law.

Keywords: mineral prospectivity mapping; Bayesian hierarchical model; spatial cross-validation; spatial heterogeneity; Central African Copperbelt; Daly tectonic domains; parameter uncertainty

1 Introduction

Mineral prospectivity mapping integrates geological, geochemical, and geophysical evidence to identify locations whose mapped characteristics are associated with known deposits. Published approaches range from knowledge-driven evidence integration (Bonham-Carter, 1994; Carranza, 2009) to logistic regression, weights of evidence (Bonham-Carter et al., 1989), Bayesian networks, and machine-learning classifiers (Porwal et al., 2006; Carranza and Laborte, 2015; Ford, 2019; Yousefi et al., 2021; Matthieu et al., 2026); they differ in assumptions, data requirements, and treatment of incomplete deposit inventories. In the Central African Copperbelt, stratigraphy, structures, basin evolution, and ore assemblages vary regionally, while the timing and detailed genesis of mineralization remain debated (Cailteux et al., 2005; Selley et al., 2005; Hitzman et al., 2012; Key et al., 2004; Ndonfack et al., 2025). The predictors used here are spatial proxies within that geological system, not direct measurements of ore-forming processes.

Spatial structure also affects evaluation. When nearby observations share covariates, occurrence history, or sampling processes, random cross-validation can place dependent observations in both

training and test sets and underestimate error for a new-region prediction task (Brenning, 2012; Roberts et al., 2017). The size and direction of this bias depend on the prediction target and blocking design; spatial blocks can also create predictor extrapolation (Valavi et al., 2019). This study therefore evaluates transfer among four contiguous along-belt sectors rather than treating random cell splitting as a proxy for geographic transfer.

The Bayesian formulation represents posterior uncertainty in shared parameters, hierarchical scales, and domain effects. Draw-wise held-out conditional probabilities are averaged to produce stored OOF scores, but the frozen OOF artifact does not retain cell-level probability intervals. Future Bernoulli outcomes are not sampled. Uncertainty in evidence layers, model parameters, conditional probabilities, and outcomes are distinct quantities in prospectivity work (Lisitsin et al., 2014; Parsa and Carranza, 2021; Zuo et al., 2021).

The Central African Copperbelt provides a setting in which to examine regional transfer. Daly’s domain labels are used as a geological stratification for model terms and secondary summaries (Daly et al., 2025); the model estimates associations indexed by those labels but does not validate the tectonic framework itself. Spatially varying coefficient models offer a statistical representation of changing relationships without making that variation causal (Gelfand et al., 2003; Fotheringham et al., 2002).

The objective is to estimate multivariate geological and geophysical predictor associations under a hierarchical Bayesian model and evaluate how fitted relationships and discrimination vary across four spatially separated Copperbelt sectors. We ask how fault distance, lithological-contact distance, Bouguer gravity, and host lithology jointly relate to the recorded occurrence labels; how model-component contributions differ across held-out sectors; how OOF discrimination varies; and whether distance-response conclusions persist under alternative mathematical representations. The uncertainty quantified in the model is parameter-posterior uncertainty; cell-level posterior probability intervals are not among the reported results.

2 Study area and data

The study area is the Central African Copperbelt tract represented by the project spatial modelling grid. The observational unit is a grid cell identified by centroid coordinates (x_i, y_i) and a binary target Y_i , where $Y_i = 1$ denotes a documented sediment-hosted stratiform copper-cobalt occurrence and $Y_i = 0$ denotes a background cell without a recorded occurrence in the compiled modeling frame. A zero is not verified evidence of barren ground. The complete-case frame contains 1,872 cells and 138 positive observations (Cailteux et al., 2005; Selley et al., 2005; Hitzman et al., 2012; Ndonfack et al., 2025).

Four continuous or categorical predictor groups are retained: distance to mapped faults, distance to mapped lithological contacts, Bouguer gravity, and host-lithology indicators. These are the predictors present in the frozen modeling data. They are spatial proxies for structural, stratigraphic, and geophysical context; their fitted coefficients are conditional predictive associations, not direct measures of causal ore-forming mechanisms. The categorical surficial lithology classification reflects the African surficial lithology parent material dataset (Sayre et al., 2014), while fault and contact distances and regional Bouguer gravity provide structural and geophysical context.

The documented positive observations are concentrated in two Daly-domain categories in this modeling frame: 107 occur in NRB_3a and 31 in NRB_3b. CRZ, NKB, SRB, and MMSB contain no

positive labels. ROC-AUC is undefined in those single-class strata; the lack of recorded occurrences does not establish geological barrenness. Copperbelt geology and mineralization are heterogeneous and multi-stage in the literature, so statistical domain effects are not interpreted as a test of a single ore-genesis model (Cailteux et al., 2005; Selley et al., 2005; Hitzman et al., 2012; Ndonfack et al., 2025).

3 Bayesian hierarchical modelling framework

3.1 Predictor preprocessing

The V11 code projects cell centroids onto the first principal component of their coordinates, sorts cells along that axis, and partitions them into four balanced contiguous groups of 468 cells. For fold k , one group is held out and the remaining groups form the training set. Spatial blocks address the intended geographic separation but do not, by themselves, eliminate all spatial dependence (Roberts et al., 2017; Valavi et al., 2019). The training and testing sets are

$$\mathcal{D}_{\text{train}}^{(k)} = \mathcal{D} \setminus B_k, \quad \mathcal{D}_{\text{test}}^{(k)} = B_k, \quad k \in \{1, 2, 3, 4\}. \quad (1)$$

Continuous predictors are standardized using only the training partition:

$$z_i = \frac{X_i - \mu_{\text{train}}}{\sigma_{\text{train}}}. \quad (2)$$

The training means and scales are applied unchanged to the held-out block. Quadratic terms are formed from standardized distances, z_f^2 and z_l^2 . Daly labels are mapped to six categories and converted to sorted integer indices. Lithology is one-hot encoded with a dropped reference category; a class is retained in a fold’s model only if it occurs among both positive and background training cells. A class found only in the held-out fold, or in only one training class, therefore has no fitted rock coefficient.

3.2 V11 model

For observation i in Daly domain $d(i)$, V11 is a partially pooled hierarchical Bayesian logistic regression (Gelman and Hill, 2007):

$$Y_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \eta_i, \quad (3)$$

where

$$\eta_i = \alpha_{d(i)} + \beta_{f,d(i)} z_{f,i} + \beta_{f^2,d(i)} z_{f,i}^2 + \beta_{l,d(i)} z_{l,i} + \beta_{l^2,d(i)} z_{l,i}^2 + \beta_g z_{g,i} + \mathbf{x}_{\text{rock},i}^T \boldsymbol{\beta}_{\text{rock}}. \quad (4)$$

The conditional probability of deposit presence for cell i given its predictors and model parameters is $p_i = \text{logistic}(\eta_i) = 1/(1 + \exp(-\eta_i))$. AUC is computed from continuous predicted probabilities and does not require selecting a classification threshold.

The intercept and the linear and quadratic fault-distance and lithological-contact-distance coefficients vary by Daly domain. Bouguer gravity enters through one global linear coefficient, and

retained host-lithology indicators enter through a global parameter vector. The domain-specific coefficients use a non-centred parameterization (Gelman and Hill, 2007),

$$\beta_{p,d} = \mu_p + \sigma_p \text{offset}_{p,d}, \quad (5)$$

for $p \in \{f_{lin}, f_{sq}, l_{lin}, l_{sq}\}$. The priors are weakly informative:

$$\mu_p \sim \mathcal{N}(0, 1), \quad \sigma_p \sim \text{HalfNormal}(1), \quad (6)$$

$$\text{offset}_{p,d} \sim \mathcal{N}(0, 1), \quad \mu_\alpha \sim \mathcal{N}(\text{logit}(\hat{p}_{train}), 1), \quad (7)$$

$$\sigma_\alpha \sim \text{HalfNormal}(1), \quad \beta_g \sim \mathcal{N}(0, 1), \quad (8)$$

$$\beta_{rock} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (9)$$

Posterior sampling uses MCMC via the No-U-Turn Sampler (Hoffman and Gelman, 2014) implemented in PyMC (Salvatier et al., 2016), with 2 chains $\times$ 1,500 retained draws per fold (3,000 total posterior draws per fold).

The compact M5 baseline is a global logistic regression using standardized Bouguer gravity, standardized lithological-contact distance, and its quadratic term. It omits fault distance and domain hierarchy. The comparison is descriptive because V11 and M5 differ in several architectural and predictor components.

4 Spatial validation and sensitivity design

Each spatial fold holds out one contiguous sector and generates out-of-fold predictions for all cells in that sector. This design tests transfer to a geographically separated part of the belt rather than random-cell interpolation (Roberts et al., 2017; Brenning, 2012). Predictive performance is summarized by ROC-AUC, PR-AUC, and Brier score where the metric is defined. AUC measures the probability that a randomly selected positive cell receives a higher predicted probability than a randomly selected negative cell; it does not require a classification threshold (Lobo et al., 2008). Bootstrap intervals reported in the source results are retained as uncertainty summaries of the performance metric, not of individual cell probabilities.

Daly-domain stratification is secondary and descriptive. Predictions are matched to domain labels after the four primary spatial models have been fitted. The model is not refitted by domain, and the analysis is neither Leave-One-Daly-Domain-Out validation nor a Fold $\times$ Domain cross-validation design. ROC-AUC is reported only for domains containing both positive and negative observations.

To assess whether distance-response inference depends on mathematical representation, four models form a 2×2 design:

Table 1: Distance-representation sensitivity design.

	Quadratic	Linear
Raw distance	Model A	Model C
Log distance	Model B	Model D

Models A and C use standardized raw distance; Models B and D use standardized $\log(1 + x_{\text{km}})$. The sensitivity analysis asks whether predictive association, curvature, and any stationary-point diagnostic persist across representations. It is a robustness analysis of the prospectivity framework, not the central scientific objective.

5 Results

5.1 Multivariate posterior structure

The fitted model jointly represents fault distance, lithological-contact distance, Bouguer gravity, and host lithology. In NRB_3a, lithological-contact distance has negative linear coefficient medians ranging from -0.849 to -0.451 and positive quadratic coefficient medians from $+0.293$ to $+0.334$ across folds. The posterior probability of positive lithological curvature ranges from 0.993 to 1.000 . NRB_3b also has negative lithological linear medians and positive quadratic medians, but with broader posterior uncertainty.

Fault-distance coefficients vary more strongly. In NRB_3a, linear coefficient medians range from -2.681 to -0.703 , while quadratic medians are negative in Fold 2 and positive in Folds 1, 3, and 4. The posterior probability of positive fault curvature ranges from 0.464 to 0.996 . In NRB_3b, fault linear medians range from $+0.181$ to $+1.548$ and the probability of positive curvature ranges from 0.161 to 0.708 . These results support recurring distance associations in the evaluated model outputs while also showing that their magnitude and direction are not uniform in every domain-specific posterior.

5.2 Spatial heterogeneity in predictor contributions

The variance decomposition concerns additive components of the out-of-fold linear predictor, not causal importance. The contribution shares are:

Table 2: Variance shares of decomposed out-of-fold linear-predictor components.

Fold	Fault distance	Lithology distance	Gravity	Host lithology
1	86.5%	11.1%	2.0%	0.4%
2	50.3%	23.9%	5.6%	20.3%
3	33.1%	28.2%	15.3%	23.4%
4	34.9%	17.7%	31.1%	16.3%

Fault distance dominates the decomposed predictor in Fold 1, contributing 86.5%. Its share declines to 50.3% in Fold 2 and approximately one-third in Folds 3 and 4. Fold 3 shows a distributed multi-factor pattern, while Fold 4 is characterized by co-dominance of fault distance and Bouguer gravity. These are spatially varying predictive contribution patterns. They are compatible with changing regional geological context but do not identify the physical mechanism producing the variation.

5.3 Distance-response sensitivity

The representation grid produced similar mean spatial ROC-AUC values: 0.686 for Model A, 0.673 for Model B, 0.671 for Model C, and 0.683 for Model D. Mean PR-AUC values were 0.107, 0.123, 0.130, and 0.145, respectively. Thus, the hierarchical linear and log-linear specifications were competitive with the quadratic specifications under the evaluated folds, and the log-linear model achieved the highest mean PR-AUC.

The recurring association is more stable than the precise functional form. Under raw distance, positive quadratic curvature is strongly supported for NRB_3a lithological-contact distance, whereas the evidence approaches ambiguity after log transformation. Fault curvature can also change sign under the log representation. The pre-registered robustness criteria were passed in 13 of 48 cases for curvature-probability agreement, 0 of 48 cases for stationary-point agreement, and 15 of 48 cases for posterior-support concentration. All three criteria were satisfied simultaneously in 0 of 48 cases.

The algebraic stationary point D^* is therefore treated as a model-derived diagnostic. At the population level, posterior curvature overlaps zero, with $P(\mu_{f,sq} > 0) = 0.384$ and $P(\mu_{l,sq} > 0) = 0.743$. The reported 95% posterior intervals span $[-279.36, 366.38]$ km for the fault-distance population diagnostic and $[-161.69, 152.34]$ km for the lithological-contact diagnostic. These intervals include physically impossible negative distances and values beyond the geographic support of the basin, so a belt-wide population D^* is not identifiable.

5.4 Spatial predictive transferability

V11 achieves fold-level ROC-AUC values of 0.817, 0.688, 0.676, and 0.561 in Folds 1–4, with corresponding PR-AUC values of 0.048, 0.103, 0.133, and 0.145. The pooled V11 ROC-AUC is 0.689, compared with 0.525 for M5, a pooled difference of +0.164 [95% interval +0.084 to +0.236]. The pooled comparison masks geographic divergence: V11 is stronger than M5 in Fold 3, while M5 is stronger in Fold 2; Folds 1 and 4 have overlapping bootstrap comparisons. Predictive transferability therefore varies along the belt even when pooled discrimination is favourable.

5.5 Secondary Daly-domain stratification

In NRB_3a, V11 has ROC-AUC 0.6617 compared with 0.5358 for M5. In NRB_3b, M5 has ROC-AUC 0.8669 compared with 0.5838 for V11. ROC-AUC is undefined in CRZ, NKB, SRB, and MMSB because those partitions contain no documented positive observations. These results provide regional context for the frozen spatial predictions; they do not constitute domain-specific model validation.

6 Discussion

6.1 What the model establishes

The results establish that geological and geophysical predictors contain predictive information about mineral prospectivity in the evaluated Copperbelt modelling frame. Fault distance, lithological-contact distance, Bouguer gravity, and host lithology contribute jointly to an uncertainty-aware

fitted predictor. The hierarchical structure permits domain-varying intercepts and distance effects while sharing information across Daly domains (Gelman and Hill, 2007). This is a statistical framework for heterogeneous prediction, not a causal demonstration that any individual predictor controls mineralization.

6.2 Spatially changing contributions

The strongest result is the spatial change in predictor contribution. Fault proximity is the dominant component in Fold 1, while the predictor share becomes more distributed across contacts, host lithology, and gravity in Folds 2–4. In the southern fold, gravity accounts for 31.1% of decomposed linear-predictor variance and is nearly co-dominant with fault distance. The pattern is consistent with a heterogeneous metallogenic belt in which the predictive expression of structural, stratigraphic, and regional geophysical information changes geographically (Cailteux et al., 2005; Hitzman et al., 2012; Ndonfack et al., 2025). Because the decomposition is model-based, it should not be read as a direct ranking of causal geological importance.

Spatially varying coefficient models in the broader statistical literature formalize the idea that regression relationships can change across space (Gelfand et al., 2003; Fotheringham et al., 2002). The hierarchical partial-pooling approach used here represents a discrete version of that concept, indexed by Daly domains rather than a continuous spatial field, while sharing information through population-level hyperparameters.

6.3 Association versus functional form

Distance associations recur across raw and logarithmic representations, but the exact nonlinear response is not uniquely identified. Quadratic terms do not provide a consistent spatial predictive advantage over linear terms, and curvature changes when the predictor scale changes. The appropriate conclusion is therefore not that the geological relationship is definitively linear, nor that all curvature is necessarily an artifact. Rather, the available evidence supports predictive association while leaving the precise distance-response functional form qualified and representation-sensitive.

6.4 What D^* does and does not tell us

The stationary point D^* is useful as an audit of a particular polynomial parameterization. It is not supported as an invariant geological distance, optimal exploration offset, or universal metallogenic property. The population-level ratio is especially unstable because posterior curvature mass overlaps zero. Domain- and fold-level diagnostics can still describe local model behaviour, but their representation dependence and fold instability prevent the interpretation of a single belt-wide distance.

6.5 Transferability and Daly’s framework

Spatially separated prediction reveals that pooled skill can conceal regional contrasts. V11 outperforms M5 in one evaluated sector and underperforms it in another, while fold contribution shares also change substantially. This supports treating along-belt transferability as a scientific result rather than merely a validation detail (Roberts et al., 2017).

Daly’s domains are most useful here as a principled geological stratification for partial pooling and regional interpretation (Daly et al., 2025). The results are consistent with regional heterogeneity in predictor relationships and predictive transferability. They do not support a universal invariant distance interpretation, and the analysis does not establish Daly’s tectonic hypothesis in its entirety.

6.6 Limitations

The modelling frame contains 1,872 cells and 138 documented positive observations, with no positives in four domain partitions. Four spatial folds provide a rigorous separation strategy but only a discrete approximation to a continuous along-belt gradient. The predictors are surface or near-surface proxies and do not represent all three-dimensional structural or subsurface geological information. The stored OOF artifact retains only the posterior mean of each cell’s conditional probability; cell-level posterior intervals were not part of the frozen result tables. Finally, the sensitivity and decomposition results are based on the frozen project result artifacts; no causal interpretation should be inferred from the predictive decomposition.

7 Conclusions

An uncertainty-aware hierarchical Bayesian model provides a coherent framework for mineral prospectivity mapping when prediction must transfer between spatially separated sectors of the Central African Copperbelt. In the evaluated modelling frame, fault distance, lithological-contact distance, Bouguer gravity, and host lithology provide recurring predictive information, but their contributions to the fitted out-of-fold predictor vary markedly along the belt. Fault proximity dominates one northern fold, whereas lithological, host-rock, and gravity contributions become more prominent elsewhere. Predictive discrimination also varies geographically, so pooled performance is insufficient to describe prospective transfer.

The distance sensitivity analysis shows that association is more stable than mathematical form. Linear and log-linear models remain competitive with quadratic models, while curvature and the derived D^* diagnostic depend on representation. Population-level D^* is not identifiable and cannot be interpreted as a universal geological distance. The broader contribution of Paper 1 is therefore an uncertainty-aware, spatially honest framework for studying heterogeneous geological predictor relationships, with qualified geological interpretation grounded in transferability and representation sensitivity rather than in a single model-derived optimum.

8 Data and code availability

The analysis materials are contained in the project repository. The primary V11 spatial workflow is implemented in `src/v11_spatial_stability_final.py`; the distance-representation sensitivity pipeline is in `src/run_distance_sensitivity.py`; and the population posterior reconstruction is in `src/run_v11_population_reconstruction.py`. The staged scientific synthesis is available under `paper.1/stages/`. Availability of the underlying spatial data and posterior traces is subject to data-sharing agreements.

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